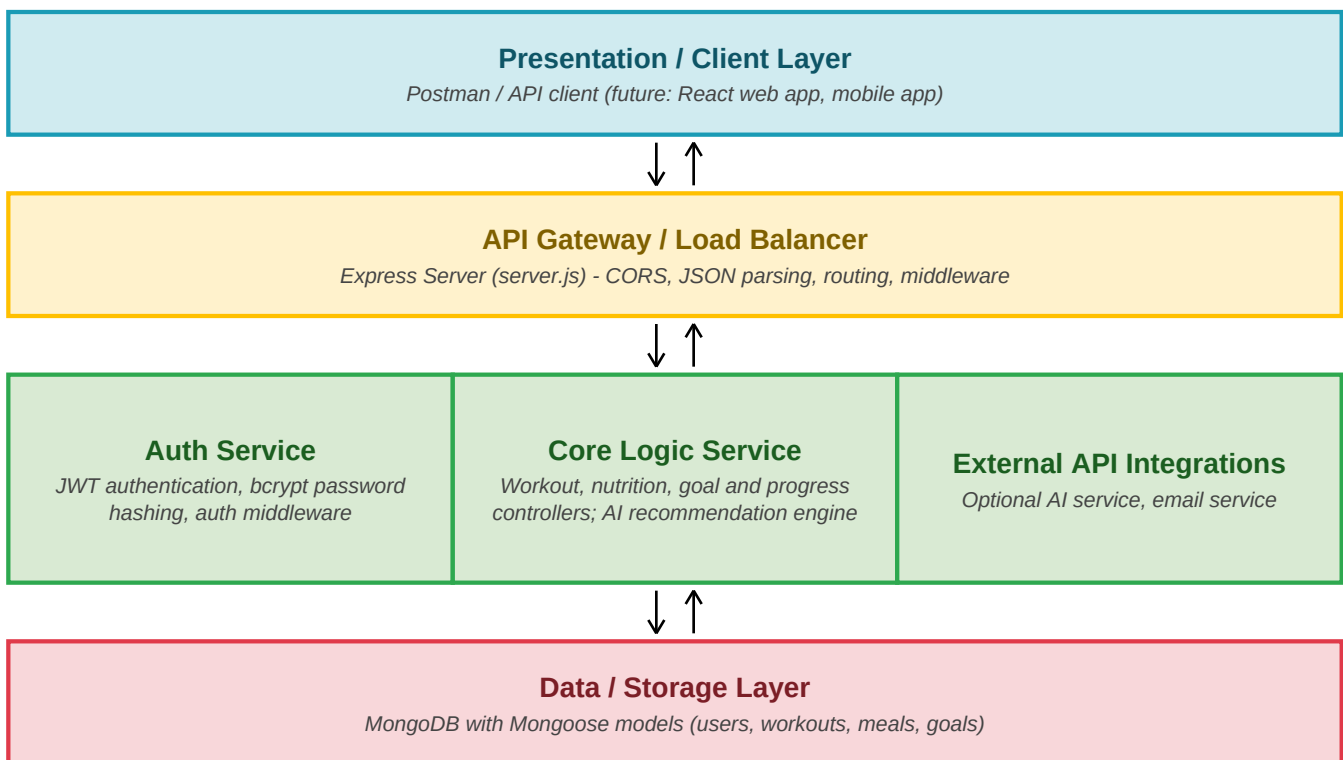


Solution Architecture

Date	08 October 2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of the AI FitTrack API showing the Presentation Layer, API Gateway, Application Services and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Sends HTTP requests to the API and displays JSON responses. Used for testing now and by future web/mobile front-ends.	Postman; (future) React / Flutter
API Gateway	Entry point of the system. Registers middleware, routes requests to controllers, and applies validation and centralised error handling.	Node.js, Express.js, CORS
Microservices / Application Services	Auth service issues and verifies JWT tokens. Core logic handles workouts, nutrition, goals and progress analytics. AI module generates workout and diet recommendations.	JWT, bcrypt, controllers, AI recommendation module
Database	Persists users, workouts, meals, goals and progress records.	MongoDB, Mongoose